

MME8215: Operations Research - Seminar

III. Transportation models.

Problem set

1. Distribution of pharma The US Department of Energy conducts a series of experiments in two research facilities located in Hawkins (Atlanta) and Los Angeles. The department needs to supply these facilities with pharma products that are processed in two factories in Portland and Flint. The delivery costs are summarised in the following table:

Production plants	Research Facilities	Delivery costs
Portland	Los Angeles	€30
Portland	Hawkins	€40
Flint	Los Angeles	€60
Flint	Hawkins	€50

The supply and demand data is provided in the table below:

Supply	Available units	Demand	Required units
Portland	200	Los Angeles	300
Flint	600	Hawkins	400

Determines how to supply the two research facilities at minimum cost.

2. Transport Planning A firm must transport machines from production plants A, B and C to warehouses X, Y and Z. Five machines are required in X, 4 in Y and 3 in Z, whereas 8 machines are available in A, 5 in B and 3 in C. The transport costs (in euros) between sites are provided in the table below.

Plant/Warehouse	X	Y	Z
A	50	60	30
B	60	40	20
C	40	70	30

1. Formulate an integer linear programming model that minimizes transport costs.
2. Assume that the cost of transporting a machine from plant B increases by €10 for all the machines as of the third one; that is, the 4th, the 5th, etc. Reformulate the model in Section a) by considering this assumption.

3. Planning production in a plastic injection company Patrick “Eel” O’Brian’s (PEOB) Ltd. is a company that manufactures plastic components for the automotive sector. As an intern at PEOB Ltd, your first assignment is to write an Integer Programming Model to calculate the optimal production plan for the plastic injection machine. The injection machine heats plastic and injects it into a mould to get a specific shape. Each mould can thus only be used to manufacture specific component types. Your first version will take into account the storage costs and the delayed orders cost which are defined below.

- *Storage costs* The storage or inventory costs represent the cost of storing the inventory levels of every component type at every planning period and are modeled as a fixed cost per unit and planning period. The storage costs are different for every component type.
- *Delayed orders costs* At every planning period, you may delay part of the demand for the next planning period. Customers will apply a fixed penalty for every delayed unit at every period. The delayed costs are also different for every component type.

Your model needs to take into account the following additional data:

- *Initial Inventory Levels (units)*: There is an initial inventory level for every component type available at the first planning period that needs to be taken into account.
- *Minimum and Maximum Inventory Levels (units)*: There is a maximum and a minimum inventory level for every component type.
- *Machine capacity (units)*: The machine capacity represents the number of units of a given component type that the machine can produce using a given mould. If a mould cannot be used to manufacture a component type, the machine capacity is zero for that combination of component type and mould.
- *Demand (Units)*: The company has several confirmed orders for the following periods and therefore, our model needs to take into account the demand for every product at every planning period.

- a. Write down the indexes and decision variables.
- b. Write down the objective function
- c. Write down the constraints

You have successfully validated your model, and it is already providing valuable information for the company. Now the company would like to extend the model to take into account as well the set-up costs:

- *Set up costs* As explained above, the injection machine uses different moulds to manufacture different parts. Each time that a mould is changed to make a different component, the operators need to set up the machine. The company estimates that this setup cost only depends on the mould that is used and is only applied when a product is changed.
- d. Modify the model to take into account the set-up costs.

4. Production Planning PLACASA, a supplier of plastic injection parts for the automotive sector, foresees the demand of an article for the following 4 months of 100, 140, 210 and 180 units, respectively. The firm can maintain only sufficient stock to meet the demand of each month, or it can have excess stock to meet the demand of two successive months or more with a storage cost of 1.2€ per month and unit of excessive stock. PLACASA calculates that the production cost for the following months, is €15, €12, €10 and €14 per unit. A preparation cost of €200 is incurred every time a production order is placed. The firm wishes to develop a production plan that minimises the total costs of production orders, of production and of keeping an article in stock.

Formulate and solve the shortest path model to find an optimum production plan.

5. Production Sequencing A manufacturer of the metal sector in Stafford manufactures four types of products in sequence on two machines. The Table below provides the necessary technical production details. Taking into account only the data on the production times per product and the maximum daily production capacity of each machine, and taking a maximum daily production of 100 units of each product type, consider and solve a maximal flow model in a graph to determine the maximum number of minutes during which machines 1 and 2 can operate on a production-in-sequence basis given the existing capacity constraints.

Machine	Product 1	Product 2	Product 3	Product 4	Daily Production Capacity (minutes)
1	2	3	4	2	500
2	3	2	1	2	380

6. Spellman's Distribution Network Optimization Spellman's would like Sabrina to focus on the optimization of their distribution network, to support the distribution of the 2 types of drinks from 3 sourcing facilities to 4 destination warehouses, through 6 intermediate warehouses. Sabrina needs to take into account the transportation costs (both from sourcing warehouses to intermediate warehouses and from intermediate warehouses to final destinations). Additionally, Sabrina can decide how many of the 6 intermediate solutions to use, given that they have a fixed operational cost that needs to be paid when the warehouse is used.

The following tables contain the transportation problem data:

Capacities of Sourcing Facilities (in units)

Sourcing Warehouse	Product A Capacity (units)	Product B Capacity (units)
S1	300	400
S2	250	150
S3	350	250

Note that the distribution costs are independent of the product type.

1. Help Sabrina write an Integer Programming Problem to model the transportation problem .
2. How would a Greedy algorithm solve the problem? Motivate your response.

Demands at Final Destinations (in units)

Destination Warehouse	Product A Demand (units)	Product B Demand (units)
D1	150	150
D2	200	250
D3	175	175
D4	225	175

Sourcing to Intermediate Warehouses Transportation Costs (Euros)

From/To	I1	I2	I3	I4	I5	I6
S1	2	4	5	7	3	6
S2	3	2	6	8	4	5
S3	1	3	4	6	5	7

Intermediate to Destination Warehouses Transportation Costs (Euros)

From/To	D1	D2	D3	D4
I1	5	2	3	4
I2	4	1	5	2
I3	3	4	2	5
I4	5	3	4	1
I5	2	5	3	4
I6	4	2	1	5

Operational Costs of Intermediate Warehouses (Euros)

Warehouse	Operational Cost
I1	1000
I2	800
I3	1200
I4	1100
I5	700
I6	900